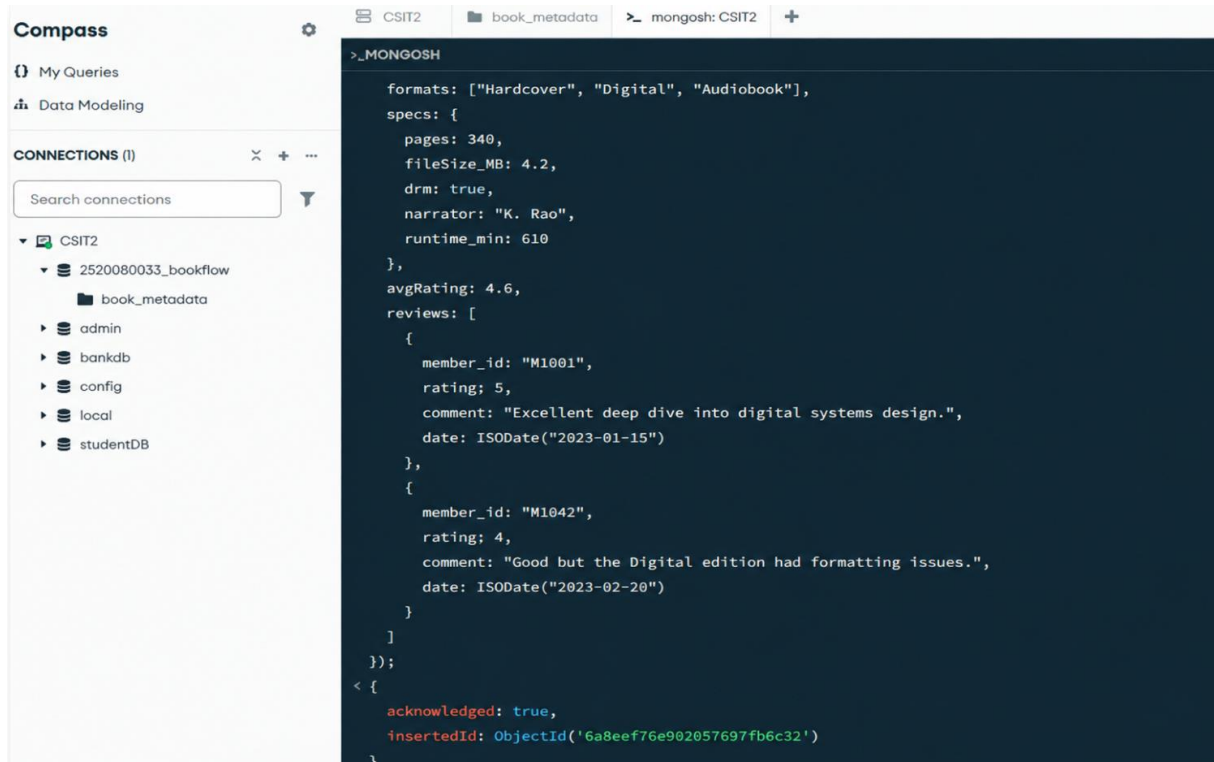


D.Shamika Hasini- 2520080033

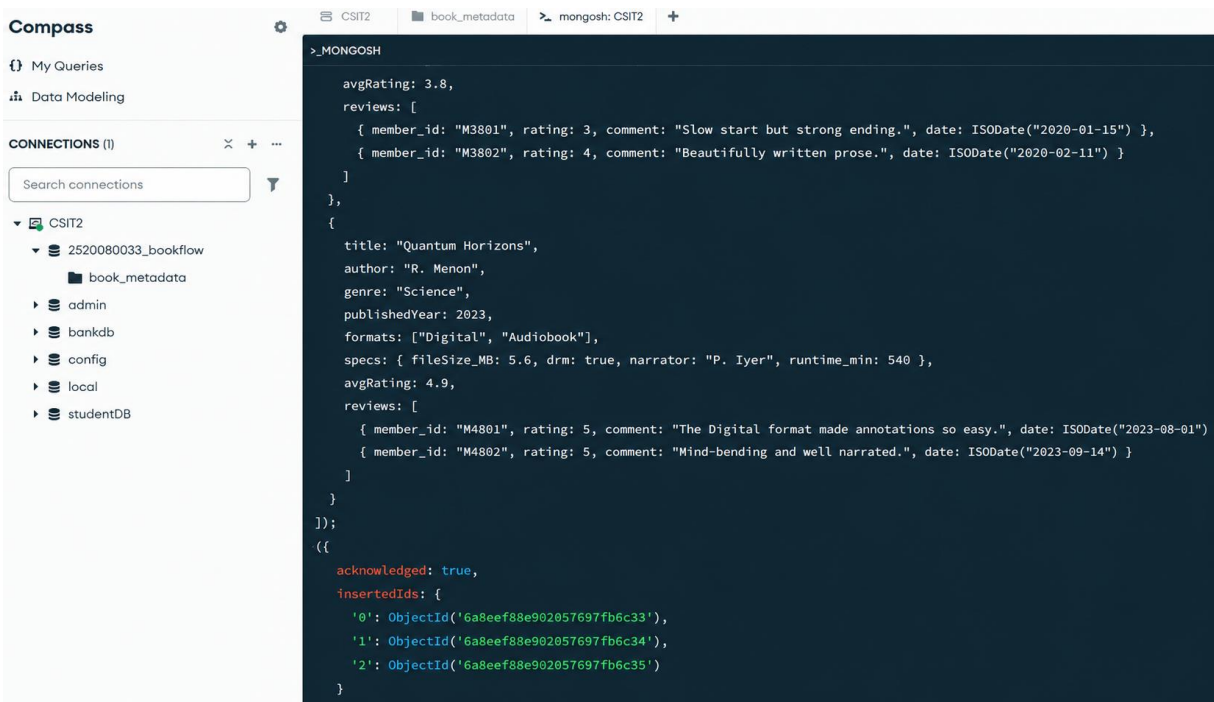
PRACTICAL WEEK-3

Task -1: Document Modelling



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'CSIT2' connection, which contains a database named '2520080033_bookflow'. Inside this database, the 'book_metadata' collection is selected. The main panel displays a document from the 'book_metadata' collection. The document contains the following fields:

```
>_MONGOSH
{
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: "K. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
}
acknowledged: true,
insertedId: ObjectId('6a8eef76e902057697fb6c32')
```



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'CSIT2' connection, which contains a database named '2520080033_bookflow'. Inside this database, the 'book_metadata' collection is selected. The main panel displays a document from the 'book_metadata' collection. The document contains the following fields:

```
>_MONGOSH
{
  avgRating: 3.8,
  reviews: [
    { member_id: "M3801", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
    { member_id: "M3802", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
  ]
},
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    { member_id: "M4801", rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") },
    { member_id: "M4802", rating: 5, comment: "Mind-bending and well narrated.", date: ISODate("2023-09-14") }
  ]
}
});
({
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8eef88e902057697fb6c33'),
    '1': ObjectId('6a8eef88e902057697fb6c34'),
    '2': ObjectId('6a8eef88e902057697fb6c35')
  }
})
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

CSIT2

2520080033_bookflow

book_metadata

admin

bankdb

config

local

studentDB

>_MONGOSH

```

avgRating: 3.8,
reviews: [
  { member_id: "M3801", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
  { member_id: "M3802", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
],
{
  title: "Quantum Horizons",
  author: "R. Menon",
  genre: "Science",
  publishedYear: 2023,
  formats: ["Digital", "Audiobook"],
  specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 },
  avgRating: 4.9,
  reviews: [
    { member_id: "M4801", rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") },
    { member_id: "M4802", rating: 5, comment: "Mind-bending and well narrated.", date: ISODate("2023-09-14") }
  ]
}
]);
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8eef88e902057697fb6c33'),
    '1': ObjectId('6a8eef88e902057697fb6c34'),
    '2': ObjectId('6a8eef88e902057697fb6c35')
  }
}

```

Task 2: Advancing query

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

CSIT2

2520080033_bookflow

book_metadata

admin

bankdb

config

local

studentDB

>_MONGOSH

```

> db.book_metadata.find({ avgRating: { $gt: 4 } });
[ {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T03:00:00.000Z
    }
  ],
}

```

1. Books rated above 4 (\$gt)

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database, which contains a 'book_metadata' collection. The main panel displays a MongoDB query in the '_MONGOSH' editor:

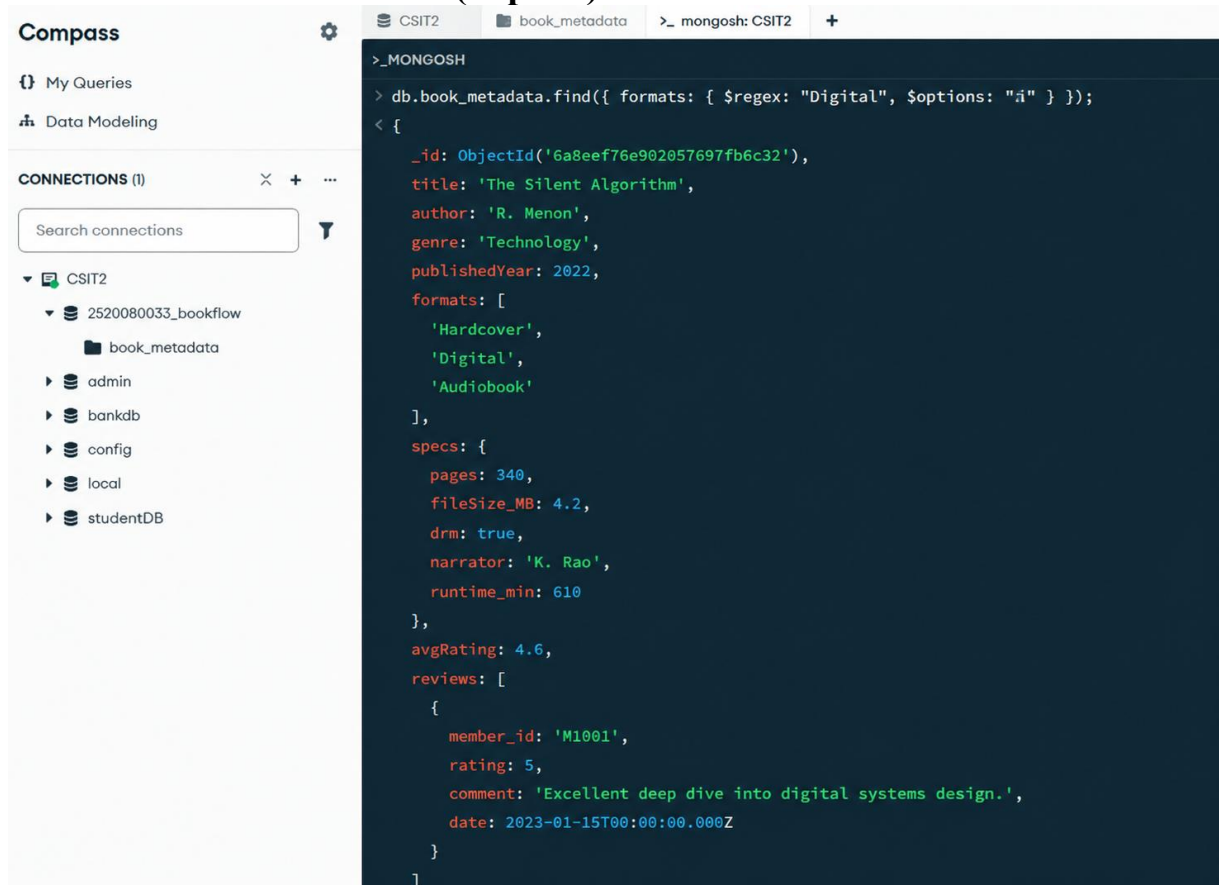
```
>_MONGOSH
> db.book_metadata.find({ avgRating: { $gt: 4 } });
< {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ],
  ...
}
```

2. Books available in specific formats (\$in)

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database, which contains a 'book_metadata' collection. The main panel displays a MongoDB query in the '_MONGOSH' editor:

```
>_MONGOSH
> db.book_metadata.find({ formats: { $in: ["Digital", "Audiobook"] } });
< {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ],
  ...
}
```

3. Combine both with Sand (implicit)

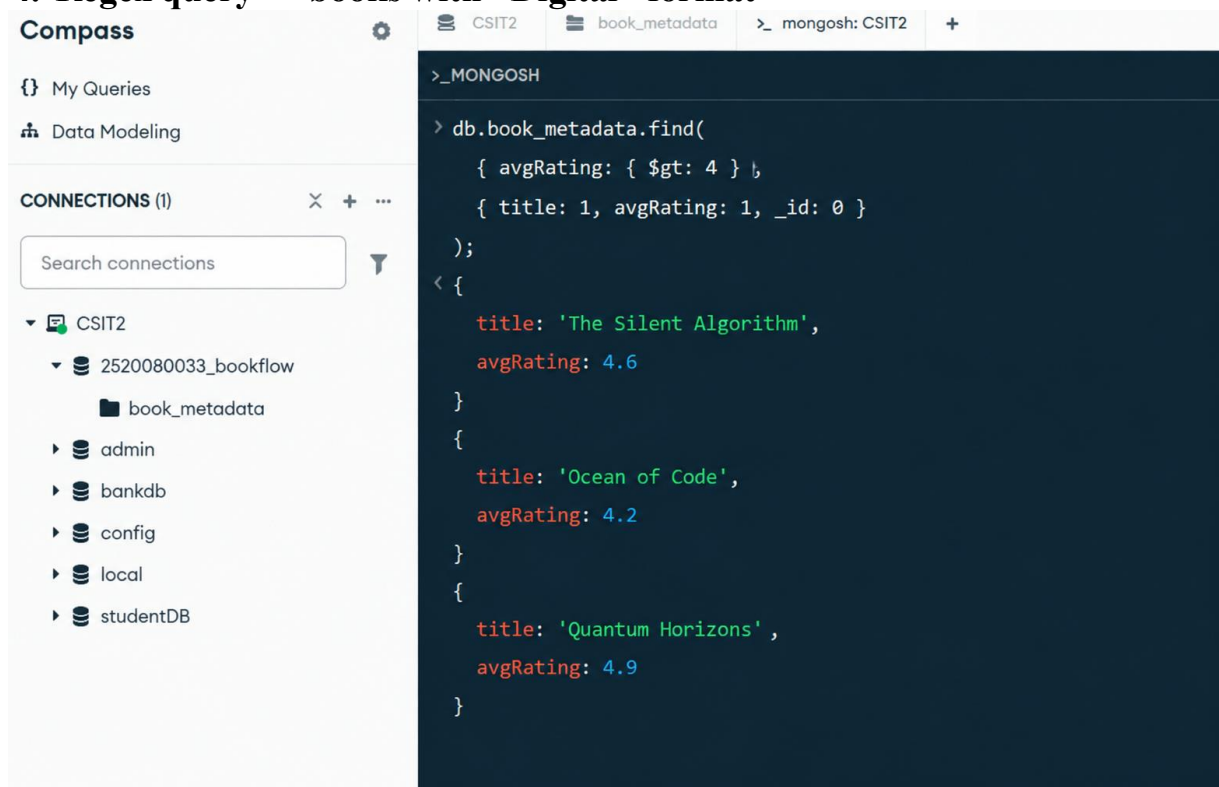


The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database and its collections, including 'book_metadata'. The main panel displays the 'MONGOSH' terminal with the following query and result:

```
> db.book_metadata.find({ formats: { $regex: "Digital", $options: "i" } });
```

```
< {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
```

4. Regex query — books with "Digital" format



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'CSIT2' database and its collections, including 'book_metadata'. The main panel displays the 'MONGOSH' terminal with the following query and result:

```
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
);
```

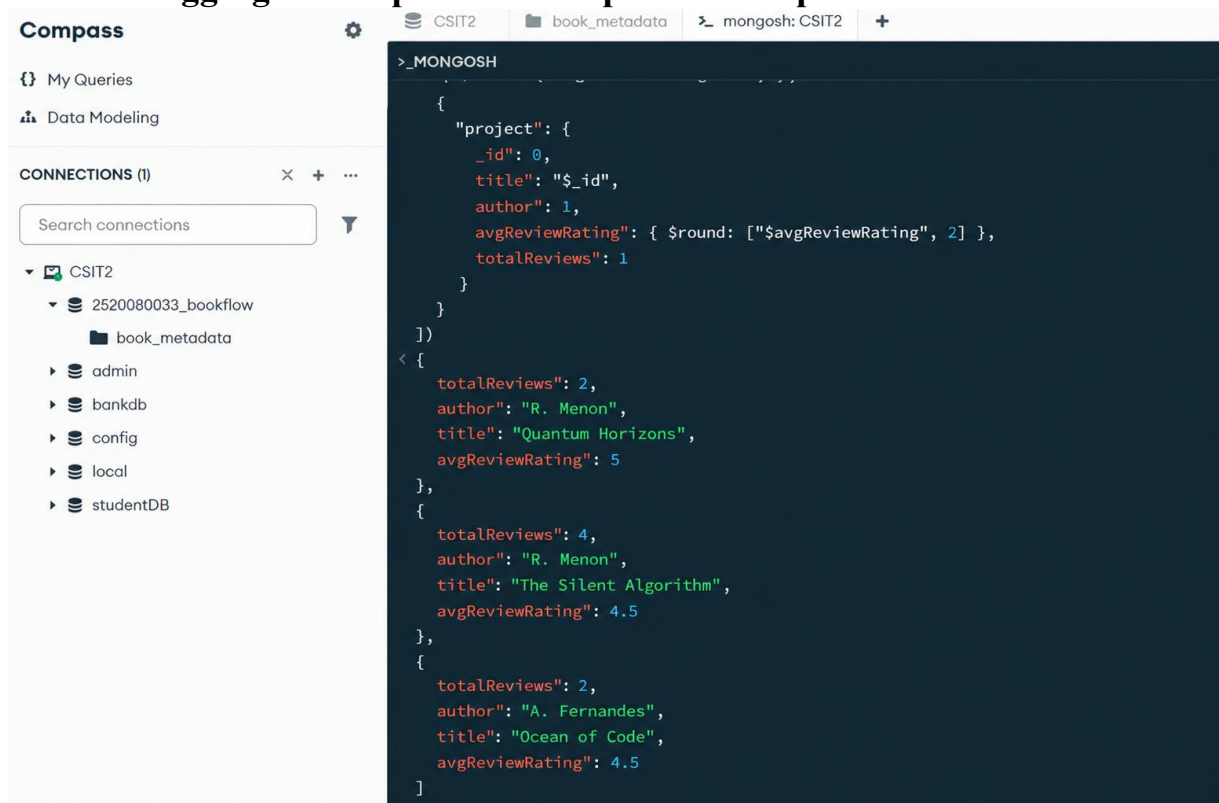
```
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
```

5. Regex query — reviews mentioning "Digital" in comments (nested field)

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel shows a connection to 'CSIT2' with a database '2520080033_bookflow' and a collection 'book_metadata'. The main panel displays a MongoDB query in the '_MONGOSH' editor:

```
>_MONGOSH
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1 }
);
< {
  _id: ObjectId('6a8eef76e902057697fb6c32'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8eef88e902057697fb6c33'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
}
```

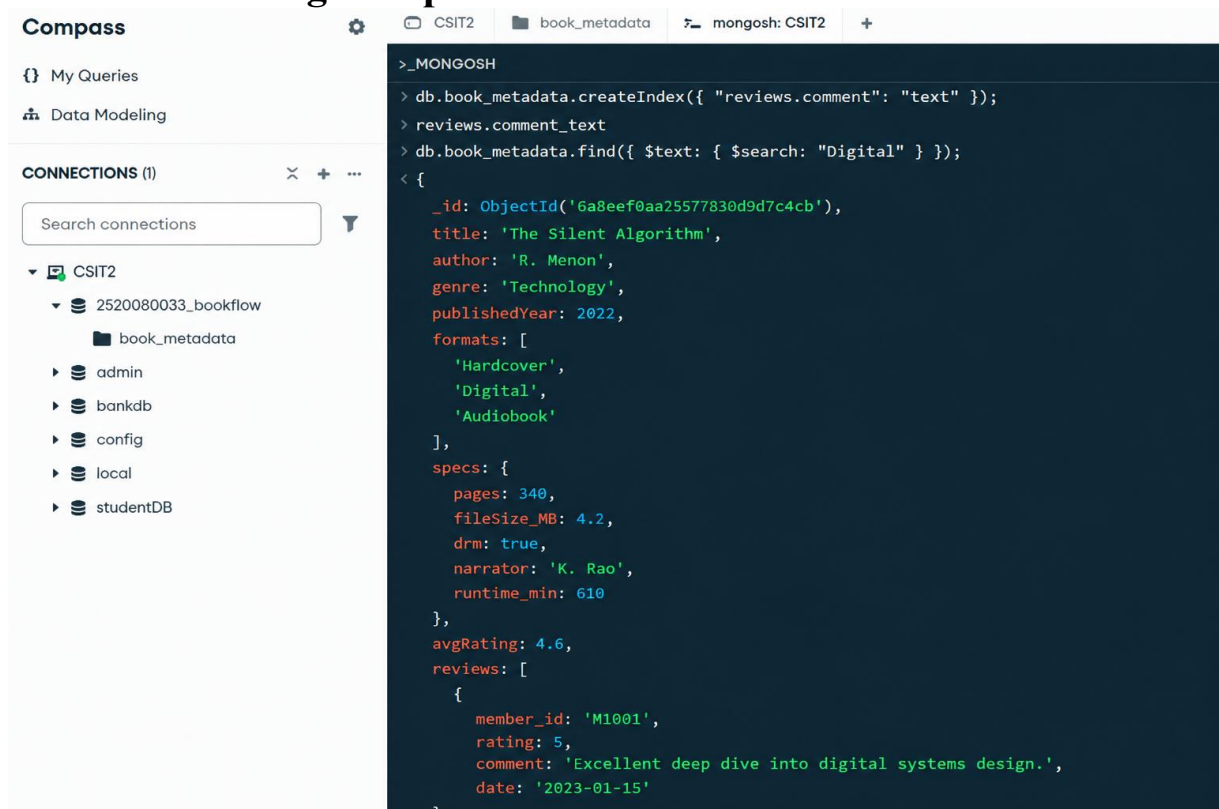
Task 3: Aggregation Pipeline — "Top Rated" Report



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'CSIT2' connection, which contains a database '2520080033_bookflow' and a collection 'book_metadata'. The main area displays the MONGOSH console with an aggregation pipeline:

```
>_MONGOSH
{
  "project": {
    "_id": 0,
    "title": "$_id",
    "author": 1,
    "avgReviewRating": { $round: [ "$avgReviewRating", 2 ] },
    "totalReviews": 1
  }
}
])
< {
  totalReviews: 2,
  author: "R. Menon",
  title: "Quantum Horizons",
  avgReviewRating: 5
},
{
  totalReviews: 4,
  author: "R. Menon",
  title: "The Silent Algorithm",
  avgReviewRating: 4.5
},
{
  totalReviews: 2,
  author: "A. Fernandes",
  title: "Ocean of Code",
  avgReviewRating: 4.5
}
]
```

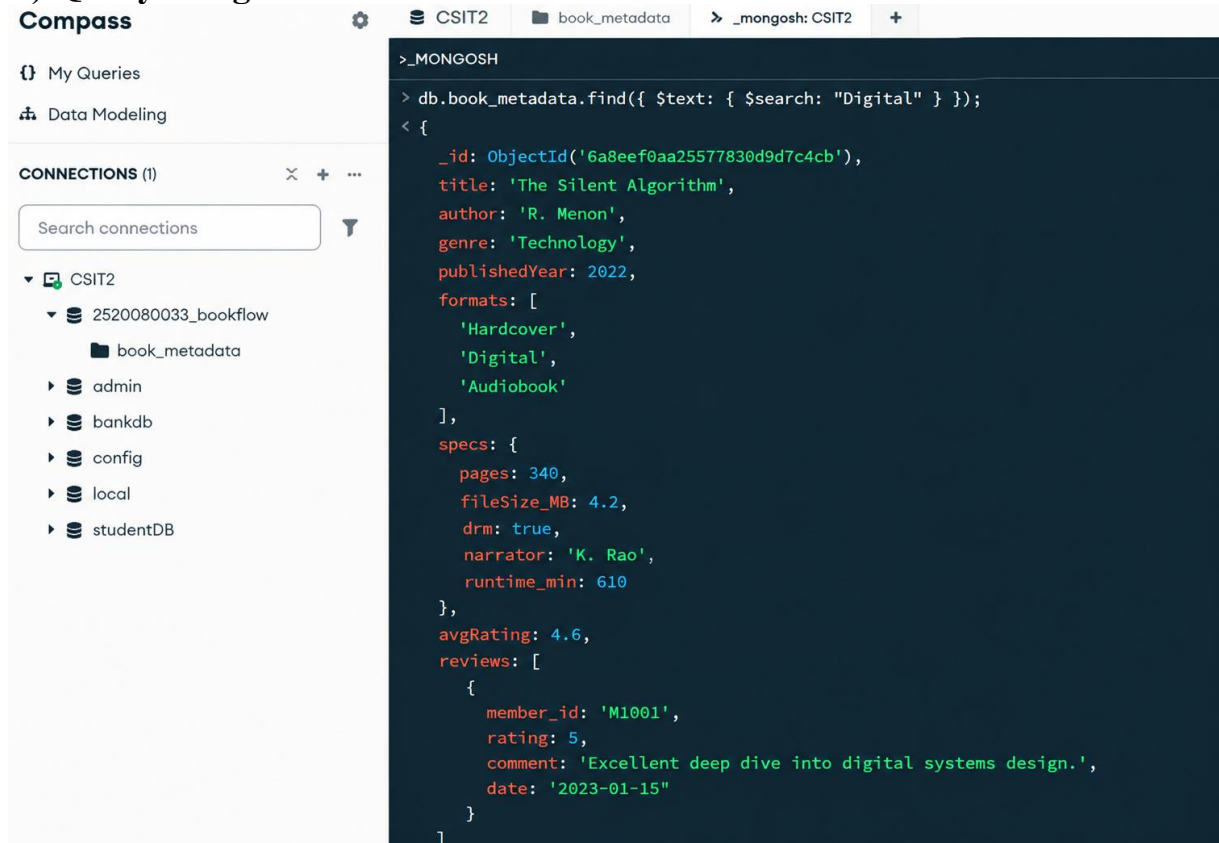
Task 4 : Indexing for Speed



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'CSIT2' connection, which contains a database '2520080033_bookflow' and a collection 'book_metadata'. The main area displays the MONGOSH console with the following commands and results:

```
>_MONGOSH
> db.book_metadata.createIndex({ "reviews.comment": "text" });
reviews.comment_text
> db.book_metadata.find({ $text: { $search: "Digital" } });
< {
  _id: ObjectId('6a8eef0aa25577830d9d7c4cb'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: '2023-01-15'
    }
  ]
}
```

b) Query using the text index



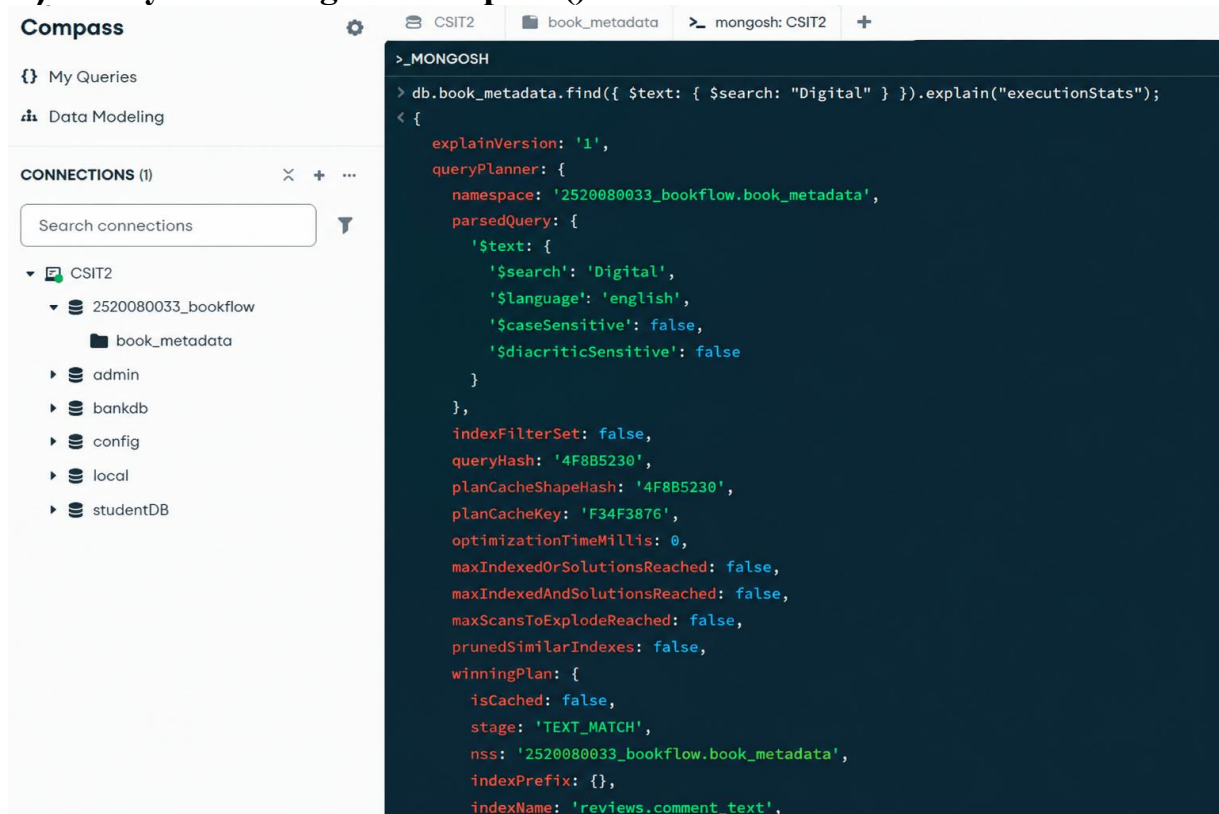
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'CSIT2' connection, expanded to show the '2520080033_bookflow' database and the 'book_metadata' collection. The main panel displays a MongoDB query in the '_MONGOSH' editor:

```
> db.book_metadata.find({ $text: { $search: "Digital" } });
```

The result is a single document:

```
< {
  _id: ObjectId('6a8eef0aa25577830d9d7c4cb'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: '2023-01-15'
    }
  ]
}
```

c) Verify index usage with .explain()



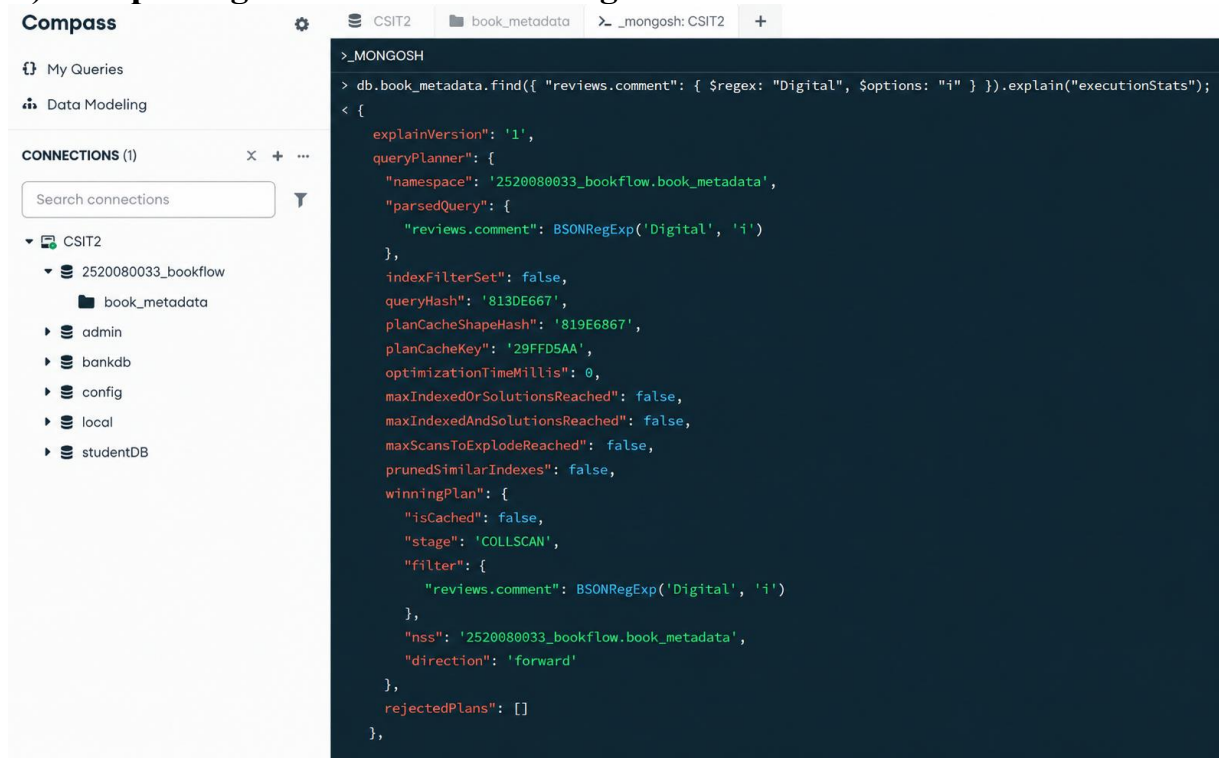
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the 'CSIT2' connection, expanded to show the '2520080033_bookflow' database and the 'book_metadata' collection. The main panel displays a MongoDB query in the '_MONGOSH' editor:

```
> db.book_metadata.find({ $text: { $search: "Digital" } }).explain("executionStats");
```

The result is a JSON object showing execution statistics:

```
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: '2520080033_bookflow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    }
  },
  indexFilterSet: false,
  queryHash: '4F8B5230',
  planCacheShapeHash: '4F8B5230',
  planCacheKey: 'F34F3876',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nss: '2520080033_bookflow.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
  }
}
```

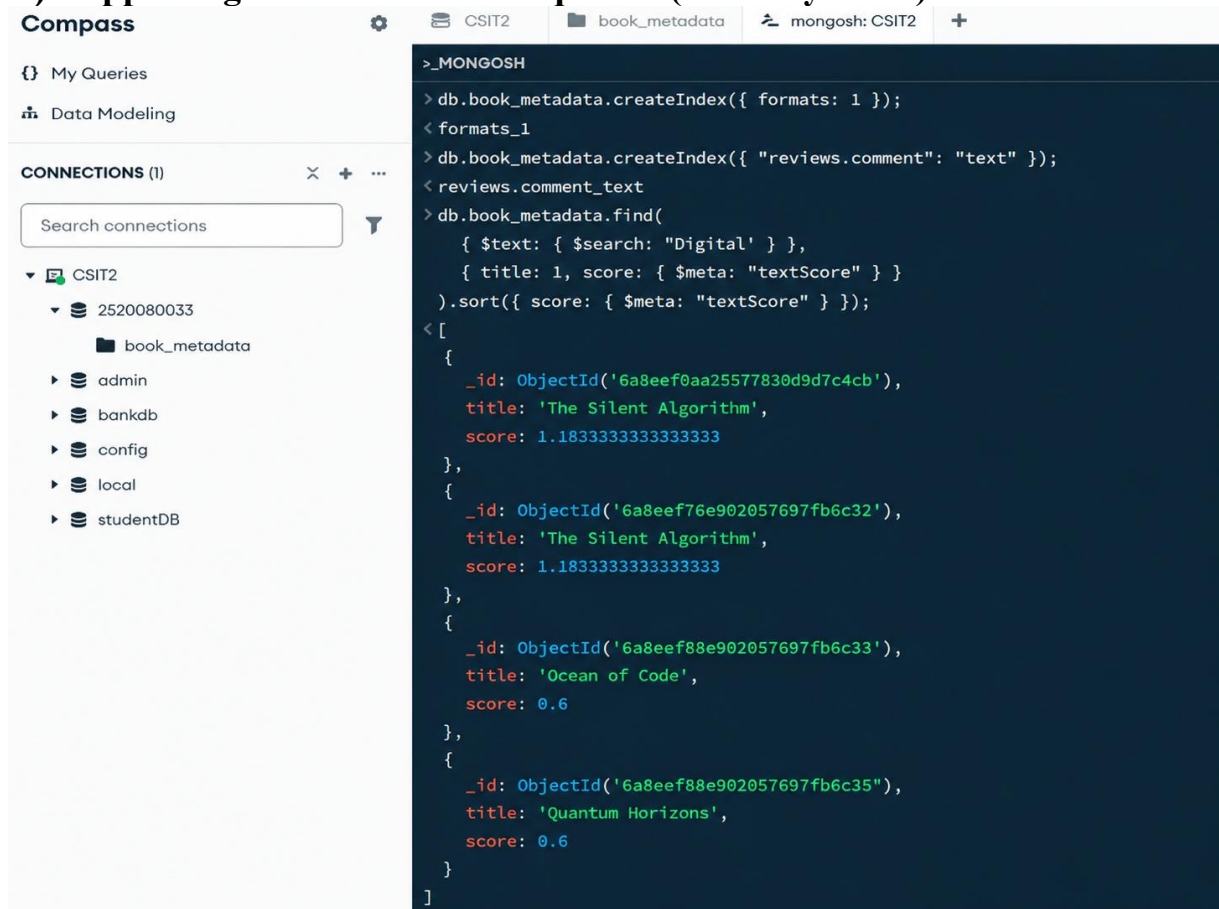
d) Compare against a non-indexed regex scan



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel shows a connection to 'CSIT2' with a database '2520080033_bookflow' and a collection 'book_metadata'. The main panel displays a query execution plan for the command: `> db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" } }).explain("executionStats");`. The plan shows a 'COLLSCAN' (collection scan) for the 'reviews.comment' field, indicating a non-indexed regex scan. The 'filter' is a 'BSONRegExp('Digital', 'i')'. The 'stage' is 'COLLSCAN' and the 'direction' is 'forward'.

```
>_MONGOSH
> db.book_metadata.find({ "reviews.comment": { $regex: "Digital", $options: "i" } }).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: '2520080033_bookflow.book_metadata',
    parsedQuery: {
      "reviews.comment": BSONRegExp('Digital', 'i')
    },
    indexFilterSet: false,
    queryHash: '813DE667',
    planCacheShapeHash: '819E6867',
    planCacheKey: '29FFD5AA',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      "isCached": false,
      "stage": 'COLLSCAN',
      "filter": {
        "reviews.comment": BSONRegExp('Digital', 'i')
      },
      "nss": '2520080033_bookflow.book_metadata',
      "direction": 'forward'
    },
    rejectedPlans: []
  },
}
```

e) Supporting index for common queries (multikey index)



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel shows a connection to 'CSIT2' with a database '2520080033' and a collection 'book_metadata'. The main panel displays a series of commands: `> db.book_metadata.createIndex({ formats: 1 });`, `< formats_1`, `> db.book_metadata.createIndex({ "reviews.comment": "text" });`, `< reviews.comment_text`, and `> db.book_metadata.find({ $text: { $search: "Digital" } }, { title: 1, score: { $meta: "textScore" } }).sort({ score: { $meta: "textScore" } });`. The results show three documents with their scores: 'The Silent Algorithm' (score: 1.1833333333333333), 'The Silent Algorithm' (score: 1.1833333333333333), and 'Ocean of Code' (score: 0.6). The last document is 'Quantum Horizons' (score: 0.6).

```
>_MONGOSH
> db.book_metadata.createIndex({ formats: 1 });
< formats_1
> db.book_metadata.createIndex({ "reviews.comment": "text" });
< reviews.comment_text
> db.book_metadata.find(
  { $text: { $search: "Digital" } },
  { title: 1, score: { $meta: "textScore" } }
).sort({ score: { $meta: "textScore" } });
< [
  {
    _id: ObjectId('6a8eef0aa25577830d9d7c4cb'),
    title: 'The Silent Algorithm',
    score: 1.1833333333333333
  },
  {
    _id: ObjectId('6a8eef76e902057697fb6c32'),
    title: 'The Silent Algorithm',
    score: 1.1833333333333333
  },
  {
    _id: ObjectId('6a8eef88e902057697fb6c33'),
    title: 'Ocean of Code',
    score: 0.6
  },
  {
    _id: ObjectId('6a8eef88e902057697fb6c35'),
    title: 'Quantum Horizons',
    score: 0.6
  }
]
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

CSIT2

- 2520080033
 - book_metadata
- admin
- bankdb
- config
- local
- studentDB

>_MONGOSH

```
explainVersion: '1',
queryPlanner: {
  namespace: '2520080033_bookflow.book_metadata',
  parsedQuery: {
    '$text': {
      '$search': 'Digital',
      '$language': 'english',
      '$caseSensitive': false,
      '$diacriticSensitive': false
    }
  },
  indexFilterSet: false,
  queryHash: '4FB85230',
  planCacheShapeHash: '4FB85230',
  planCacheKey: 'F34F3876',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nss: '2520080033_bookflow.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [
```

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

CSIT2

- 2520080033_bookflow
 - book_metadata
- admin
- bankdb
- config
- local
- studentDB

>_MONGOSH

```
:db.book_metadata.find({ title: 'The Silent Algorithm' })
:[
  {
    _id: ObjectId('6a8eef0aa25577830d9d7c4cb'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [
      'Hardcover',
      'Digital',
      'Audiobook'
    ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: '2023-01-15'
      }
    ]
  }
]
```